# HOSPITAL MANAGEMENT SYSTEM

# Modern UI / UX Design Specification

Production-oriented design system, screen architecture, workflows and responsive UX


Design direction: Calm clinical intelligence — enterprise SaaS efficiency, strong information hierarchy, fast workflows and trustworthy visual language.

This document is derived from the supplied HMS project scope/PRD and Master TRD, then expanded with current healthcare UI/UX research.

# 1. Product UX Strategy

The HMS is an operational ERP rather than a public marketing site. The interface must help receptionists register and move patients quickly, doctors review clinical context safely, accountants reconcile financial records accurately, pharmacists act on inventory signals, laboratory staff process samples and reports, franchise partners understand earnings, and administrators monitor the whole hospital.

## 1.1 Core UX Goals

- Reduce unnecessary clicks in high-frequency workflows.
- Make operational state immediately scannable: waiting, admitted, occupied, available, low stock, pending report, unpaid and pending withdrawal.
- Use progressive disclosure so complex data remains manageable.
- Keep clinical information visually calm and separate from financial/administrative emphasis.
- Make destructive and financial actions explicit.
- Keep interaction patterns consistent across every role.

## 1.2 Role-Based Experience Model

| Role | Primary UX Goal |
|---|---|
| Super Admin | See hospital-wide performance, configuration, security and exceptions. |
| Reception | Move patients through registration, appointment, token, OPD/IPD and billing quickly. |
| Doctor | Focus on today's patients, history, prescription, reports and follow-ups. |
| Accountant | Reconcile invoices, payments, cash/bank books, ledgers and P&L. |
| Pharmacist | Find medicine quickly, manage stock/expiry and complete billing. |
| Lab Technician | Process test requests, samples and report generation with minimal friction. |
| Franchise Partner | Understand referrals, commissions, wallet balance and withdrawals. |

The supplied quotation explicitly includes these web panels and the Doctor Android application; it also states that a patient mobile application is not included in the package. filecite turn1file0 L22-L44

# 2. Visual Design System

## 2.1 Design Personality

| Principle | Visual Treatment |
|---|---|
| Trust | Deep navy/blue anchors, predictable controls, clear system states. |
| Calm | Light neutral workspace, restrained color and generous spacing. |
| Professional | Enterprise SaaS layout, precise tables and strong typography. |
| Fast | Dense but scannable operational views and prominent contextual actions. |
| Human | Friendly empty states, useful microcopy and non-alarming errors. |

## 2.2 Color Tokens

| Token | Use | Value |
|---|---|---|
| Navy | Navigation / headings | #123B5D |
| Action Blue | Primary CTA / active / links | #1E6FD9 |
| Clinical Teal | Secondary clinical accent | #16A6B6 |
| Success | Available / completed / paid | #159A67 |
| Warning | Low stock / pending / attention | #D58B00 |
| Danger | Critical / cancelled / destructive | #D64545 |
| Canvas | Application background | #F5F8FB |
| Text | Primary text | #17212B |
| Muted | Secondary text | #667684 |

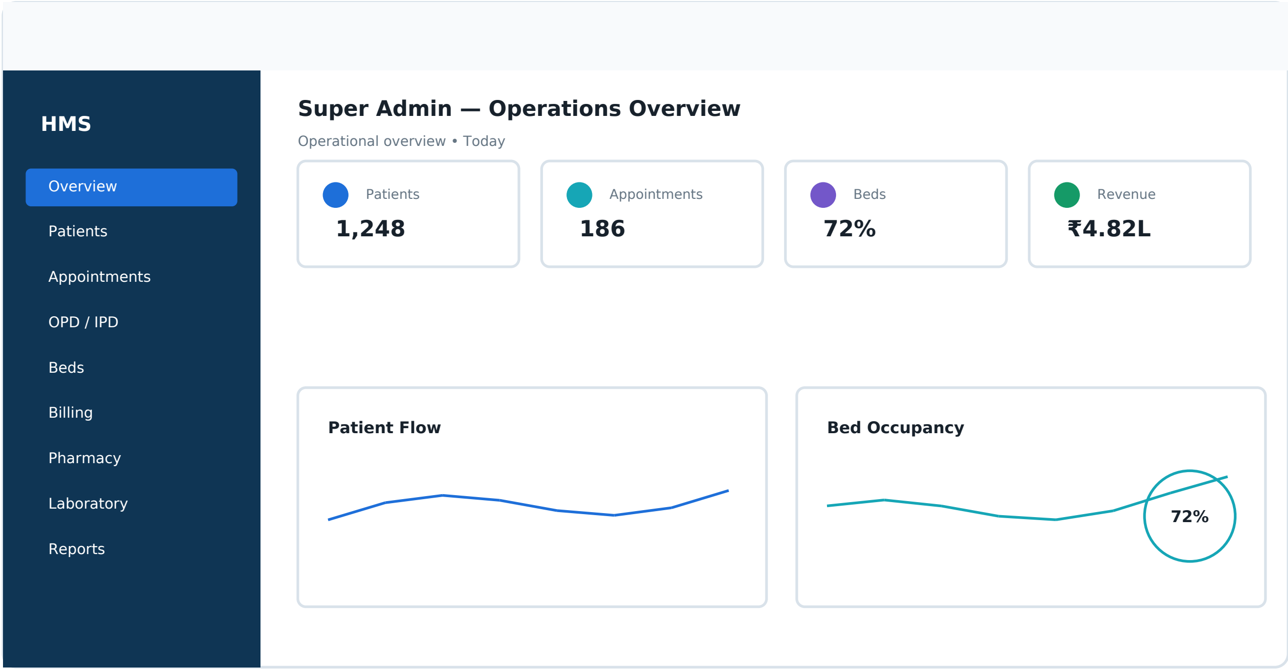
| Token | Use | Value |
|---|---|---|
| Border | Dividers / controls | #D9E2EA |

Use color semantically. Red should indicate a genuine risk or destructive action; green should communicate a meaningful positive state, not decoration.

### 2.3 Typography & Spacing

| Element | Specification |
|---|---|
| Display | 28–32 px / 700 |
| Page title | 22–24 px / 700 |
| Section heading | 17–18 px / 700 |
| Body | 14–15 px / 400–500 |
| Caption | 12–13 px / 400–500 |
| KPI numbers | 24–30 px / 700 |
| Spacing | 8 px base rhythm: 8 / 16 / 24 / 32 / 40 |
| Radius | Cards 12–16 px; inputs 8–10 px; modals 16 px |

## 3. Application Shell & Information Architecture




### 3.1 Global Shell

| Region | UX Specification |
|---|---|
| Sidebar | Collapsible desktop navigation; role-filtered modules; active state; actionable count badges. |
| Top bar | Global search, notifications, help, role/profile menu. |
| Page header | Breadcrumb, title, context, primary action and optional date/filter controls. |
| Workspace | 12-column responsive grid for dashboards, forms and data views. |
| Quick actions | Contextual actions such as Register Patient, New Appointment, New Invoice. |
| Command search | Patient, UHID, token, invoice and navigation search. |

### 3.2 Role Navigation

| Role | Primary Navigation |
|---|---|
| Super Admin | Overview, Patients, Doctors, Appointments, OPD/IPD, Beds, Billing, Accounting, Pharmacy, Laboratory, Inventory, Franchise, Reports, Users/Permissions, Settings, Audit Logs |
| Reception | Dashboard, Patients, Appointments, Token Queue, OPD, IPD, Beds, Billing, Documents |
| Doctor | Dashboard, Today's Appointments, Patients, Prescriptions, Reports, Admissions, Follow-ups, Earnings, Profile, Leave |
| Accountant | Dashboard, Invoices, Payments, Cash Book, Bank Book, Ledger, Expenses, Balance Sheet, P&L, Reports |
| Pharmacy | Dashboard, Inventory, Purchases, Suppliers, Expiry, Low Stock, Barcode Billing, Reports |
| Laboratory | Dashboard, Test Categories, Requests, Samples, Report Queue, PDF Reports, Billing, Reports |
| Franchise | Dashboard, Referrals, Patients Referred, Wallet, Withdrawals, Earnings, Analytics, Profile |

# 4. Dashboard UX

The dashboard must answer three questions immediately: What is happening now? What needs attention? What should I do next?

## 4.1 Executive Dashboard Zones

| Zone | Content |
|---|---|
| KPI strip | Patients, appointments, occupied beds, collections/revenue, pending reports, low-stock alerts. |
| Patient flow | Today vs previous period; OPD/IPD volume; waiting/completed/cancelled. |
| Bed utilization | ICU, General, Private and Deluxe occupancy + availability. |
| Appointments | Next appointments, token queue and status. |
| Financial pulse | Collections, outstanding, expenses and revenue trend. |
| Operational alerts | Low stock, expiry, pending reports, pending withdrawals. |
| Quick actions | Patient, appointment, admission, invoice, lab request, pharmacy bill. |

## 4.2 Dashboard Interaction Rules

- Every KPI card should deep-link to the filtered module.
- Charts provide date/filter controls and readable hover/tap detail.
- Alerts are actionable, not informational dead ends.
- Default dashboard state is today with explicit date controls.
- Skeleton loading preserves layout and avoids content jumps.

# 5. Patient Management UX

## 5.1 Patient List

| Element | Design |
|---|---|
| Search | UHID / name / phone; search field receives immediate keyboard focus. |
| Filters | Doctor, department, visit type, admitted/active and date. |
| Table | UHID, patient, phone, age/gender, latest visit, status and actions. |
| Actions | View, edit, appointment, OPD/IPD, invoice, documents. |
| Responsive | Desktop table; mobile prioritized patient cards. |

## 5.2 Patient Profile

- Header: patient name, UHID, age/gender, contact, status and contextual actions.
- Tabs: Overview, Medical History, Visits, Prescriptions, Lab Reports, IPD Admissions, Documents, Billing.
- Timeline view for previous visits and relevant clinical events.
- Clinical records and financial records use distinct visual grouping.
- Show timestamps and document/report status.

## 5.3 Registration Flow

- Two-column desktop form; single-column mobile form.
- Mandatory demographic fields are above the fold.
- After creation, show UHID immediately and provide next actions.
- Recommended next actions: Book Appointment / Start OPD / Admit to IPD.
- Use duplicate-patient warning patterns based on strong matching information.

# 6. Appointment & Token UX

## 6.1 Appointment Views

| View | Purpose |
|---|---|
| Day | Reception/doctor operational view with token order and time blocks. |
| Week | Doctor availability and scheduling conflicts. |
| List | Fast filtering, status management and operational reporting. |

## 6.2 Appointment Card

- Patient name + UHID.
- Doctor + department.
- Time + token number.
- Type: OPD / Walk-in / Online.
- Status: Scheduled / Waiting / In Consultation / Completed / Cancelled / No-show.
- Context actions based on current user permission.

## 6.3 Token Queue

- Large current token with high contrast.
- Next 5–10 tokens visible.
- Doctor/department filter.
- Waiting-time indicator.
- Clear actions for call/advance/consultation state.
- Status never depends on color alone.

# 7. OPD / IPD / Bed Management UX

## 7.1 OPD Workspace

| Area | Pattern |
|---|---|
| Queue | Token-based patient list with status chips. |
| Patient context | Compact patient summary pinned beside consultation workflow. |
| Clinical notes | Structured sections with expandable detail. |
| Prescription | Medicine rows with dosage, duration and instructions. |
| Follow-up | Date, notes and next action. |

## 7.2 IPD Admission

- Admission header shows patient, admission date and current status.
- Bed selection uses a visual availability matrix.
- Available / Occupied / Maintenance are visually distinct.
- Transfer confirmation explicitly shows old bed → new bed.
- Discharge summary gets a focused workflow rather than a crowded general form.

## 7.3 Bed Visual System

| State | Visual treatment | Primary action |
|---|---|---|
| Available | Soft green indicator + neutral card | Allocate |
| Occupied | Blue/navy indicator + patient context | View / Transfer |
| Maintenance | Amber/red indicator + lock icon | Manage |

The supplied project explicitly includes ICU, General Ward, Private and Deluxe bed categories, availability, allocation, transfers and occupancy reporting. filecite turn1file0 L77-L86

# 8. Billing & Accounting UX

## 8.1 Billing Layout

| Region | Design |
|---|---|
| Invoice header | Invoice number, patient name/UHID, date, module. |
| Item area | Line items with quantity, unit price, discount and applicable tax. |
| Summary panel | Subtotal → discount → GST → paid → due → total. |
| Payment | Payment mode, amount and confirmation. |
| Actions | Save draft / Print / PDF; destructive actions require confirmation. |

## 8.2 Accountant Dashboard

| Widget | Content |
|---|---|
| Collections | Today / week / month |
| Outstanding | Amount + aging |
| Cash Book | Opening, inflow, outflow, closing |
| Bank Book | Transactions and reconciliation |
| P&L | Revenue, expense and net result |
| Ledger | Recent entries with filters |

Billing requirements in the supplied scope include OPD, IPD, Pharmacy, Laboratory and Package billing, GST invoices, receipts, payment history and outstanding reports. filecitetturn1file0L87-L105

# 9. Pharmacy & Laboratory UX

## 9.1 Pharmacy

- Search-first interface with barcode field visually prioritized.
- Inventory table: medicine, batch where available, quantity, expiry, purchase price, selling price and status.
- Low-stock and expiry are one-click operational filters.
- Stock status is visible without opening a detail page.
- Barcode billing supports rapid scanner/keyboard input.

## 9.2 Laboratory

- Request queue grouped by Pending Sample / Collected / Processing / Ready.
- Sample collection screen optimized for fast patient/request selection.
- Report screen separates patient/test metadata from result content.
- PDF generation status is visible with retry on failure.
- Ready reports provide clear preview, download and print actions.

## 9.3 Inventory & Purchase

- Master-detail purchase workflow.
- Supplier/vendor context remains visible during purchase entry.
- Stock movement history is accessible from item detail.
- Expiry warnings are filterable and reportable.

# 10. Franchise / Channel Partner UX

The partner portal should be intentionally simpler than the full ERP. Its mental model is: referrals → successful services → earned commission → wallet → withdrawal.

## 10.1 Partner Dashboard

| Area | Design |
|---|---|
| Referral KPI | Patients referred, successful referrals, conversion. |
| Earnings | Current period + lifetime earnings. |
| Wallet | Available and pending balance. |
| Withdrawals | Pending / approved / paid history. |
| Analytics | Doctor/service/surgery referral performance where authorized. |

## 10.2 Commission Transparency

- Each commission line identifies its source referral/bill where privacy rules permit.
- Commission rule is explicit: Fixed, Percentage, Doctor-wise, Service-wise or Surgery-wise.
- Wallet movements appear as a ledger, not only a balance.
- Withdrawal state and timestamps are visible.

The supplied scope includes referral tracking, commission wallet, withdrawal requests, earnings, doctor/franchise referrals and multiple commission rule types. fileciteturn1file0L128-L153

# 11. Doctor Android App UX

The mobile client should not reproduce the whole desktop ERP. It should optimize for one-handed, rapid clinical review.

| Tab | Purpose |
|---|---|
| Home | Today's appointments, pending items and quick actions |
| Appointments | Today's/upcoming schedule |
| Patients | Search and recent patients |
| Reports | Medical/lab reports |
| Profile | Profile, leave, earnings and settings |

## 11.1 Doctor Home

- Doctor identity + notification control.
- Today's appointment count and next patient.
- Upcoming appointment cards with token/time.
- Quick actions for patient, prescription and report.
- Follow-ups, pending reports and earnings summary.

## 11.2 Prescription UX

- Patient identity pinned at top.
- Diagnosis and notes visually separated.
- Medicine search/autocomplete with compact structured fields.
- Dosage, duration and instructions remain readable on small screens.
- Save action is persistent and returns explicit success/failure state.

The supplied scope specifies secure doctor login, today's appointments, patient history, digital prescriptions, medical reports, admission details, follow-up, notifications, earnings and profile management. fileciteturn1file0L32-L44

# 12. Component Design System

| Component | Rule |
|---|---|
| Button | Primary / Secondary / Ghost / Danger; one dominant action per context. |
| Input | Label above, helper/error below, visible focus state. |
| Select | Searchable for doctors, patients, medicines and tests. |
| Table | Sticky header where useful, pagination, filtering and responsive fallback. |
| Status Chip | Semantic state + icon/text; never color alone. |
| Modal | Confirmation/short task; avoid complex workflows inside small modals. |
| Drawer | Quick detail/edit while preserving list context. |
| Toast | Short confirmation; actionable errors include navigation/retry. |
| Tabs | Use for patient/profile domains and preserve context. |
| Empty State | Explain why empty + primary next action. |
| Skeleton | Mirror final layout to prevent layout shift. |

## 12.1 Data Visualization

- Prefer simple line, bar, donut and occupancy visuals.
- Do not chart data when a number/table is clearer.
- Provide accessible text summaries for charts.
- Keep date ranges and legends consistent.
- Financial charts clearly distinguish revenue, expense and outstanding.

# 13. UX States, Accessibility & Responsive Design

| State | Requirement |
|---|---|
| Loading | Skeleton/spinner near affected content; preserve layout. |
| Empty | Explain state and provide next action. |
| Error | Human-readable message + retry/correction. |
| Success | Explicit confirmation and resulting record/status. |
| Forbidden | Explain access limitation without exposing protected data. |
| Network | Show connection problem and safe retry; never claim a failed write succeeded. |
| Conflict | Explain changed state and offer refresh/review. |

## 13.1 Accessibility

- Target WCAG 2.2 AA principles for contrast, keyboard access, focus and semantics.
- Do not communicate status by color alone.
- All controls have visible labels.
- Keyboard navigation must support reception/accounting workflows.
- Touch targets must be comfortably tappable.
- Charts/icons require text equivalents.
- Focus remains predictable after modals and validation.

## 13.2 Responsive Strategy

| Width | Strategy |
|---|---|
| ≥1440 px | Full sidebar, 12-column dashboard and dense tables. |
| 1024–1439 px | Collapsible sidebar, 8/12-column adaptive grid. |
| 768–1023 px | Tablet two-column forms; dense tables may scroll horizontally. |
| <768 px | Single-column, card-based lists and task-first mobile layout. |

# 14. Critical End-to-End UX Flows

| Flow | UX sequence |
|---|---|
| New Patient → OPD | Register → UHID → Appointment → Token → Waiting → Consultation → Prescription → Billing → Complete |
| IPD | Patient → Admission → Bed Matrix → Allocate → Treatment → Transfer if needed → Billing → Discharge → Bed Released |
| Pharmacy | Scan/Search → Stock/Expiry Check → Items → Bill → Payment → Inventory Deduction → Receipt |
| Laboratory | Doctor Request → Billing → Sample → Processing → Report PDF → Ready → View/Print |
| Referral | Referral → Service/Bill → Rule Evaluation → Commission → Wallet → Withdrawal → Settlement |

## 14.1 Interaction Rules

- Every workflow exposes current state.
- Every major transition provides immediate feedback.
- Users never need to infer whether an API action succeeded.
- Identifiers remain visible: UHID, token, invoice number, admission and lab request.
- Use confirmation only for meaningful consequences to avoid confirmation fatigue.

# 15. Contemporary UI/UX Research

Recent healthcare dashboard references were reviewed to identify broad 2025–2026 patterns rather than copy a particular product. HavenMed emphasizes clean patient, appointment, staff, medical-record, billing and inventory management with real-time insights. CareSync emphasizes real-time patient, appointment, revenue and departmental performance visibility. Mediva emphasizes patient flow, bed utilization, financial performance and service metrics. citeturn0search0turn0search1turn0search12

Other current healthcare concepts reinforce a calm clinical palette, structured dashboards, accessible information hierarchy and mobile-first operational views. citeturn0search16turn0search18turn0search19

## 15.1 Patterns Adopted

- Calm blue/teal clinical language.
- KPI-first dashboard followed by actionable detail.
- Patient and appointment views optimized for rapid scanning.
- Visual bed occupancy.
- Light cards, restrained borders/shadows and rounded corners.
- Responsive/mobile-first thinking for clinical workflows.

## 15.2 Patterns Explicitly Avoided

- Decorative glassmorphism that reduces readability.
- Neon or excessive gradient styling.
- Dashboard overload with unrelated charts.
- Tiny typography in operational tables.
- Color-only status communication.
- Copying any reference product's visual identity or layout.

# 16. Vue.js Design-to-Code Mapping

| Design System | Implementation Direction |
|---|---|
| Tokens | Central CSS variables/theme tokens. |
| Components | Reusable component library with documented variants. |
| Shell | Responsive AppShell + role-aware navigation. |
| State | Pinia for auth, permissions and domain state. |
| Routing | Vue Router + permission guards. |

| Design System | Implementation Direction |
| --- | --- |
| API | Axios service layer + standard response/error handling. |
| Tables | Reusable filtering/pagination/sort patterns. |
| Forms | Reusable fields + consistent validation. |
| Charts | One accessible charting system. |

### 16.1 Recommended Component Naming

AppShell → Sidebar → Topbar → PageHeader → DashboardGrid → StatCard / DataCard → DataTable → FormSection → Modal/Drawer → Toast/Alert. Domain components should use explicit names such as PatientSummaryCard, AppointmentQueue, BedStatusGrid, InvoiceSummary, LabRequestQueue and CommissionWallet.

### 16.2 Engineering Rule

Create each interaction pattern once and reuse it. Do not create visually similar but behaviorally different buttons, inputs, status chips or tables for each module unless the behavior genuinely differs.

## 17. Security-Aware UX

The project scope includes RBAC, secure REST APIs, encrypted database handling, SSL, daily backups, audit logs and Firebase notifications. filecite turn1file0 L166-L173 The UI should make these controls understandable without exposing implementation secrets.

- Show current user role in the profile/session area.
- Hide unauthorized navigation items where appropriate, while retaining server-side authorization.
- Use confirmation for high-impact actions.
- Mask sensitive values according to role policy.
- Never display API tokens or security credentials.
- Provide clear session-expiration/re-authentication messaging.

## 18. UI/UX Acceptance Checklist

- Every web role has role-appropriate navigation and dashboard priorities.
- Patient registration is fast and does not require unnecessary navigation.
- Appointment/token workflow is optimized for reception.
- OPD/IPD and bed states are immediately scannable.
- Billing has a clear financial hierarchy.
- Pharmacy supports scan/search-first interaction.
- Laboratory queues make sample/report status obvious.
- Franchise users understand referral → commission → wallet → withdrawal.
- Doctor Android focuses on today's clinical tasks.
- Every major screen has loading, empty, error and success states.
- Responsive behavior is explicitly designed.
- Keyboard, focus and contrast are considered.
- Components are reusable and implementation-ready for Vue.js.
- No critical status relies on color alone.
- Critical workflows provide explicit completion feedback.

## 19. Final Design Direction

Product feel: Calm clinical intelligence. The HMS should look like a modern enterprise healthcare SaaS platform: restrained, precise, fast and trustworthy. Sophistication comes from hierarchy, spacing, typography, data visualization and interaction quality—not decoration.

Desktop: Navy/blue navigation + light neutral workspace + white cards + semantic teal/green/amber/red states + readable data tables.

Mobile: Focused cards, large touch targets, persistent context, minimal navigation and task-first clinical workflows.

Implementation priority: Build the design system and shell first, then Patient → Appointment → OPD/IPD → Beds → Billing → Pharmacy → Laboratory → Franchise/Reports → Android Doctor App. This sequence supports consistent UI and the project's 25-working-day implementation target.

# Appendix — Source & Research Notes

The supplied project quotation identifies the HMS as a centralized hospital ERP covering OPD/IPD, bed allocation, appointments, doctors, pharmacy, laboratory, billing, accounting, franchise/channel partner commissions, inventory and reporting. fileciteturn1file0L16-L21

The quotation identifies responsive web panels for Super Admin, Doctor, Reception, Accountant, Pharmacy, Laboratory and Franchise/Channel Partner users. fileciteturn1file0L22-L31

The scope also includes pharmacy, laboratory and inventory functions that directly drive the screen architecture in this design specification. fileciteturn1file0L106-L127

## External references reviewed

HavenMed — Hospital Management Dashboard UI/UX Design, Behance, published February 2025. citeturn0search0

CareSync — Healthcare Admin Dashboard UI Design, Behance, published February 2026. citeturn0search1

Mediva — Hospital Management Dashboard and Patient List Dashboard, Dribbble. citeturn0search12turn0search15

Healthcare Admin Dashboard UI/UX Design — Medical SaaS Web App, Dribbble. citeturn0search18turn0search21

Hospital Management Enterprise Web Design, Dribbble. citeturn0search19

These references informed broad contemporary patterns only. No external design was reproduced as-is.